

Computational Fluid Dynamics II

CFD Lab 1: Direct numerical simulation (DNS) of turbulent channel flow at $Re\tau \approx 180$

Due date: 18 May 2025

Deliverables: Group report + codes

Report format: Typed/hand-written

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Task 1:

SIMSON is an incompressible boundary layer solver for boundary layer flows. It uses spectral techniques to solve the Navier-Stokes equations.

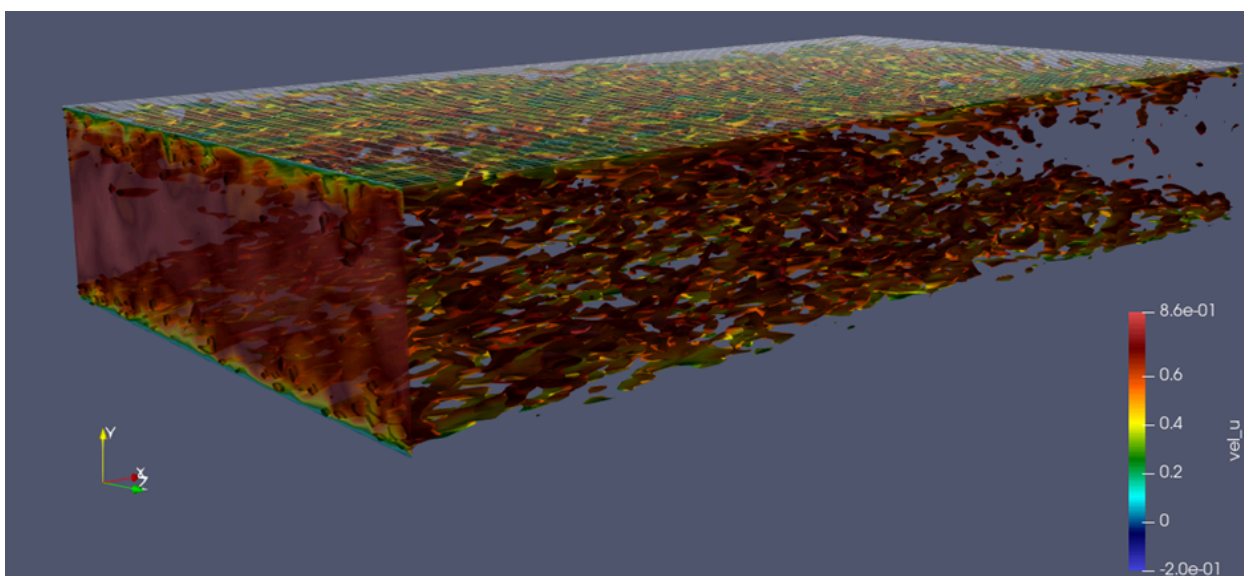
Procedure:

- The Simulation was run for Re_b of 4200 for a wall bounded channel flow with 64,65,64 for n_x, n_y, n_z respectively.
- The simulation was run using MPI for 8 threads. For one run, the simulation was run with a dynamic dt and for another run, it was run with a static dt of 0.02

Visualization of instantaneous vortical structures in channel flow:

The lambda 2 method helps to detect vortex structures. We observed that the more we decrease lambda 2, the vortices captured decreased i.e. more structures at -0.5 than -5, indicating that only very strong vortices are captured if we use say -10. If the lambda is positive, no vortices are seen.

The front wall has been plotted as per the fluctuations in U velocity. It is clear that the wall effects are well seen in the pseudo colour plot. We also observe the y grid is stretched, with more cells at the wall.



Task 2:

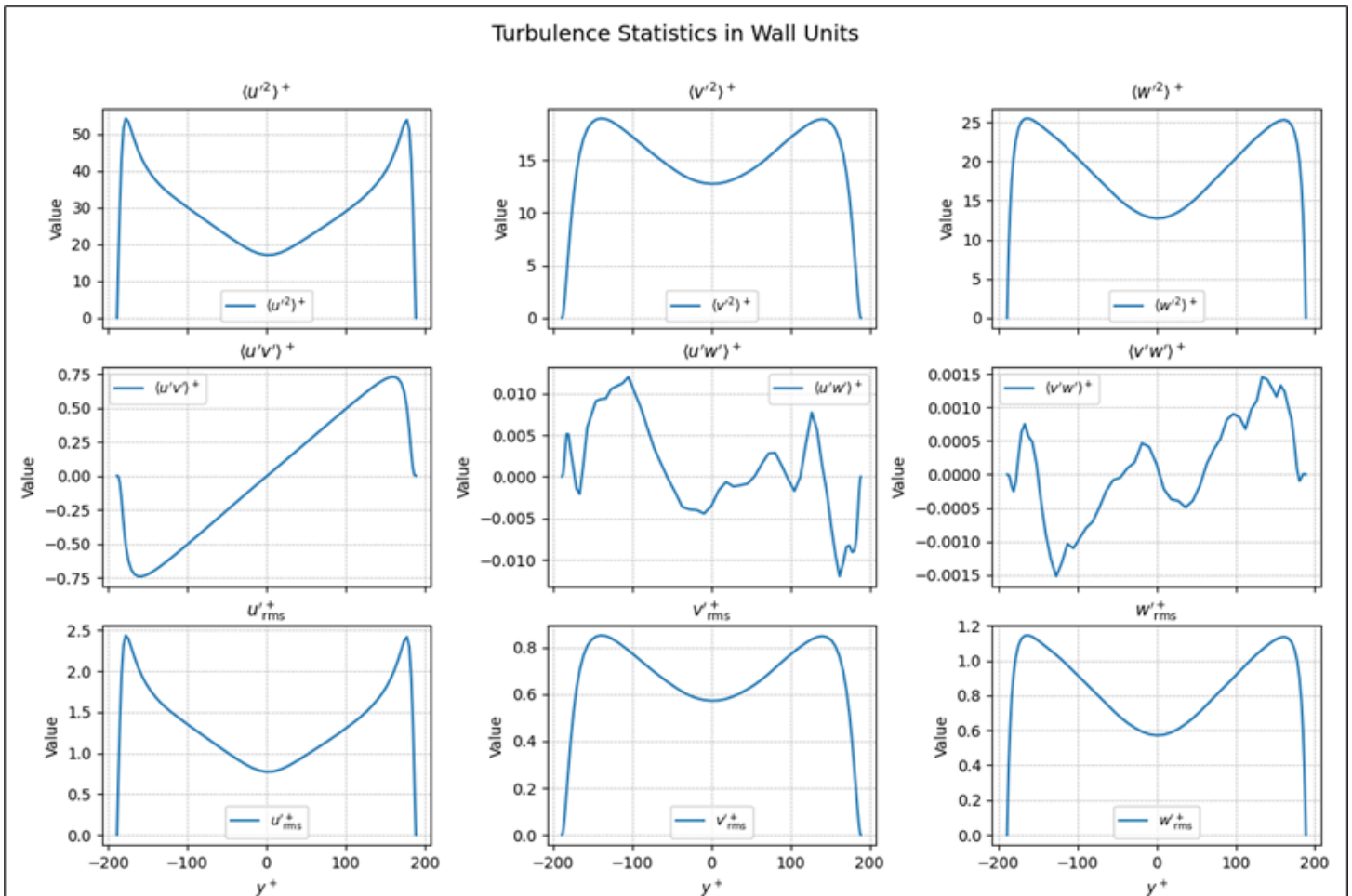
The rms values were sourced directly from the stat.vel output file created by SIMSON.

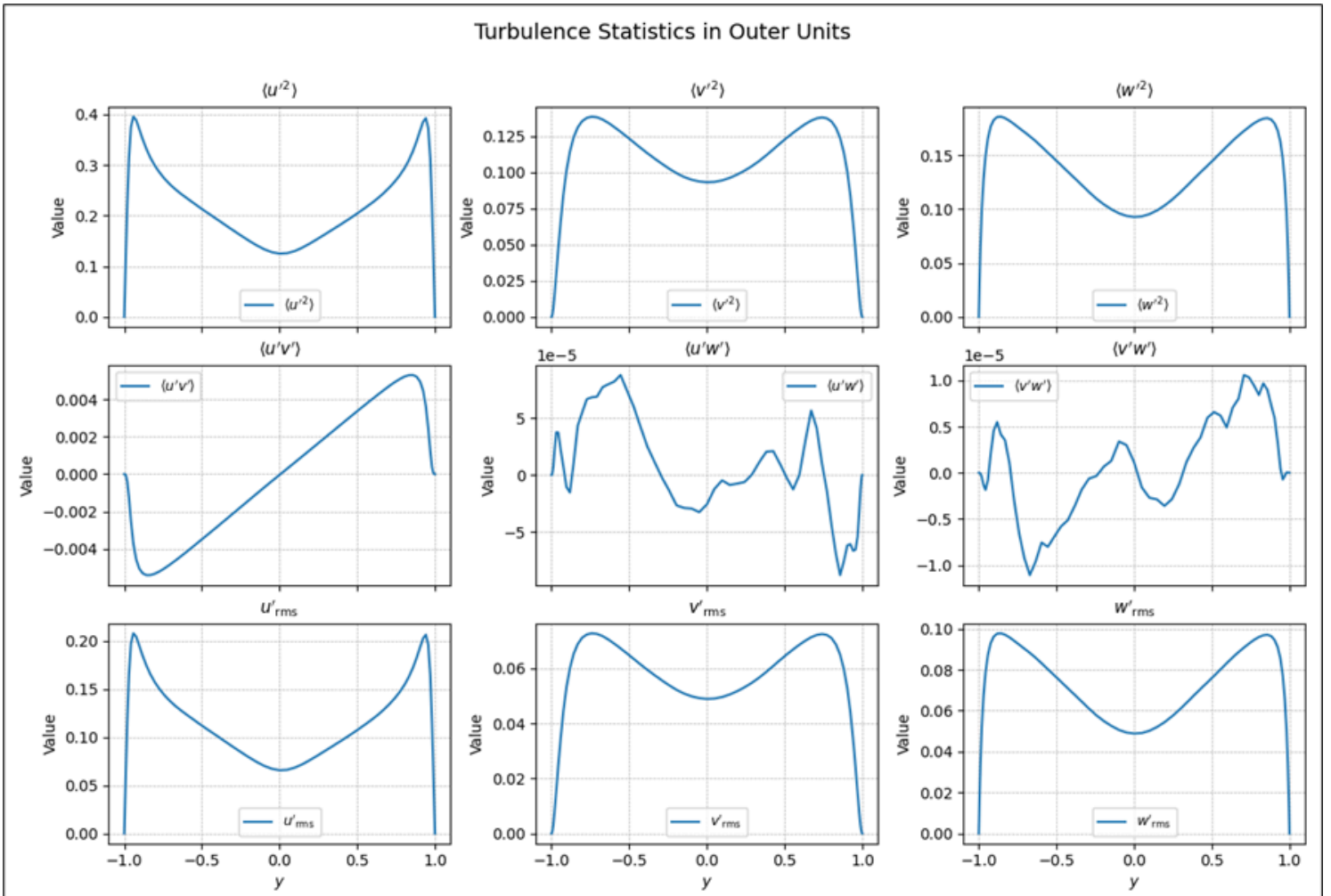
The fluctuations can be obtained by squaring these. For plus units, the quantities were non dimensionalised by (u_τ) and y is normalised by l^* . The gradients required are calculated using the reference code chebdf which allows us to account for stretched grid.

For outer units, the quantities are non dimensionalised by U_b and H for stress and y coordinates respectively. The bulk velocity can be calculated using the centreline velocity

and is given as
$$U_b = \frac{2}{3} \cdot U_0$$

We can see that the profiles for outer and inner units are similar in nature, with difference in scaling. The turbulent shear stresses are much greater near the wall, and lesser at the centreline ($y = y^+ = 0$). (Note: The code works correctly when the full folder is opened through VS Code).





Task 3:

Transport equation for turbulent kinetic energy

$$C = P + \Pi + D + T - \varepsilon \quad k = \frac{1}{2} \langle u'_i u'_i \rangle$$

$$C = U_j \frac{1}{2} \frac{\partial u'_i u'_i}{\partial x_j} \quad : \text{convection (}=0 \text{ in parallel flows)}$$

$$P = -\langle u'_i u'_j \rangle \frac{\partial U_i}{\partial x_j} \quad : \text{Production, Adds energy to fluctuations (mainly u)}$$

$$\Pi = \frac{\partial \langle u'_j p' \rangle}{\partial x_j} \quad : \text{Pressure transport, Redistributes between directions}$$

$$D = 2\nu \frac{\partial \langle u'_i s'_{ij} \rangle}{\partial x_j} \quad : \text{Viscous Diffusion}$$

$$T = -\frac{1}{2} \frac{\partial \langle u'_i u'_i u'_j \rangle}{\partial x_j} \quad : \text{Turbulent diffusion} \quad s'_{ij} = \frac{1}{2} \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)$$

$$\varepsilon = 2\nu \langle s'_{ij} s'_{ij} \rangle \quad : \text{Dissipation}$$

M = mass L = length T = Time

$$C = U_j \frac{1}{2} \frac{\partial u'_i u'_i}{\partial x_j} = \frac{(\text{velocity})^3}{(\text{length})} = \frac{(m/s)^3}{m} = m^2 s^{-3} = M^0 L^2 T^{-3}$$

To non-dimensionalize the convection term we need to divide with the same quantity ($L^2 T^{-3}$)

$$u_T^x (l^*)^y = M^0 L^2 T^{-3}$$

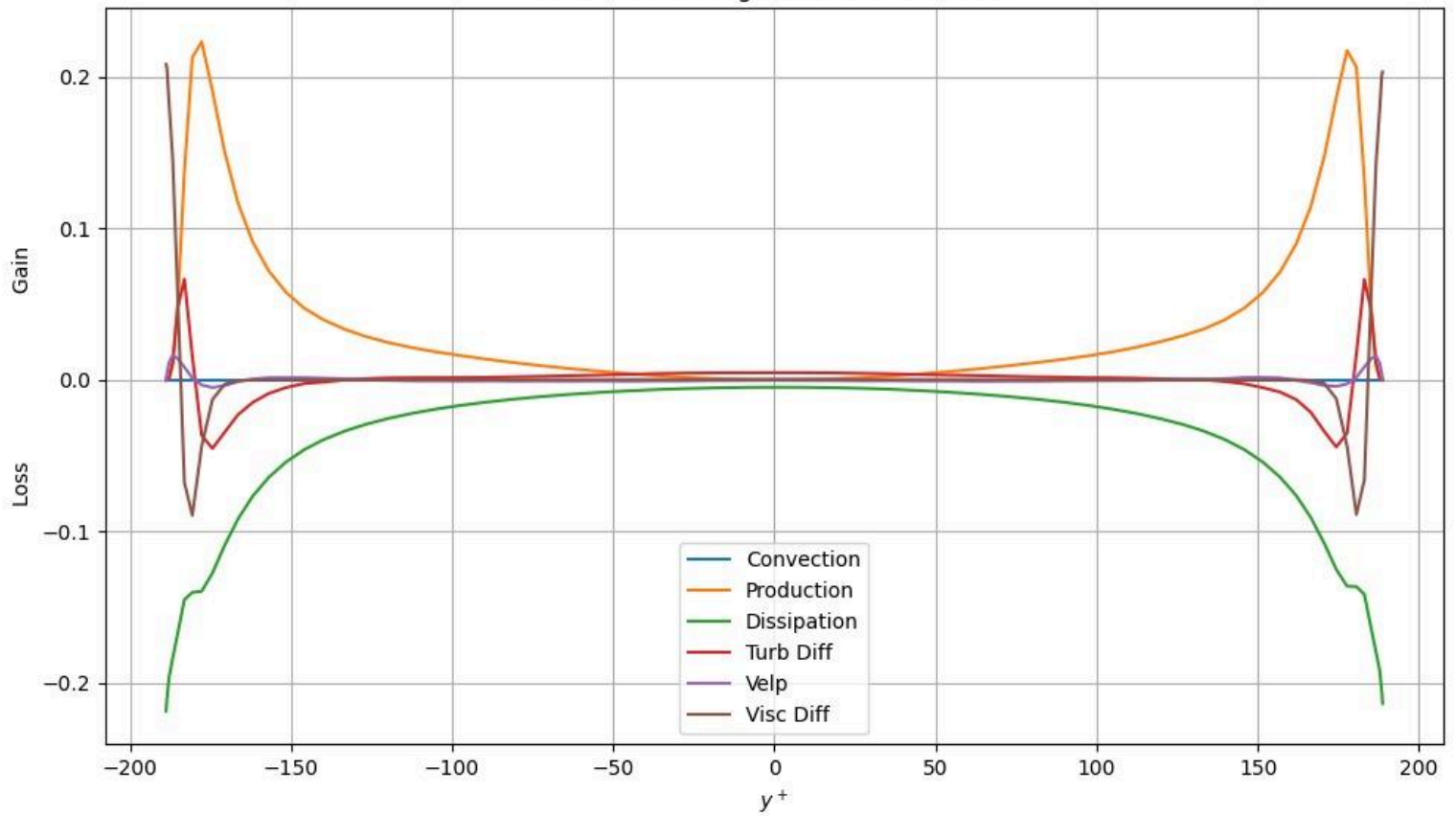
$$x = 3 \quad \text{and} \quad y = -1$$

$$[u_T^x (l^*)^y] = [u_T^3 (l^*)^{-1}] = L^2 T^{-3}$$

$$C^+ = \frac{C}{u_T^3 (l^*)^{-1}} = \frac{L^2 T^{-3}}{L^2 T^{-3}} = 1$$

The same process is done for remaining terms in transport equation for turbulent kinetic energy.

Inner scaled Budget Terms in Wall Units



Task 4:

$$u_{T_1} = \sqrt{\nu \left| \frac{du}{dy} \right|} \quad \text{at top wall}$$

$$u_{T_2} = \sqrt{\frac{1}{Re} \left| \frac{du}{dy} \right|} \quad \text{at bottom wall}$$

$$Re_{T_1} = u_{T_1} Re \quad Re_{T_2} = u_{T_2} Re$$

Average friction Reynolds number is

$$Re_T = \frac{Re_{T_1} + Re_{T_2}}{2}$$

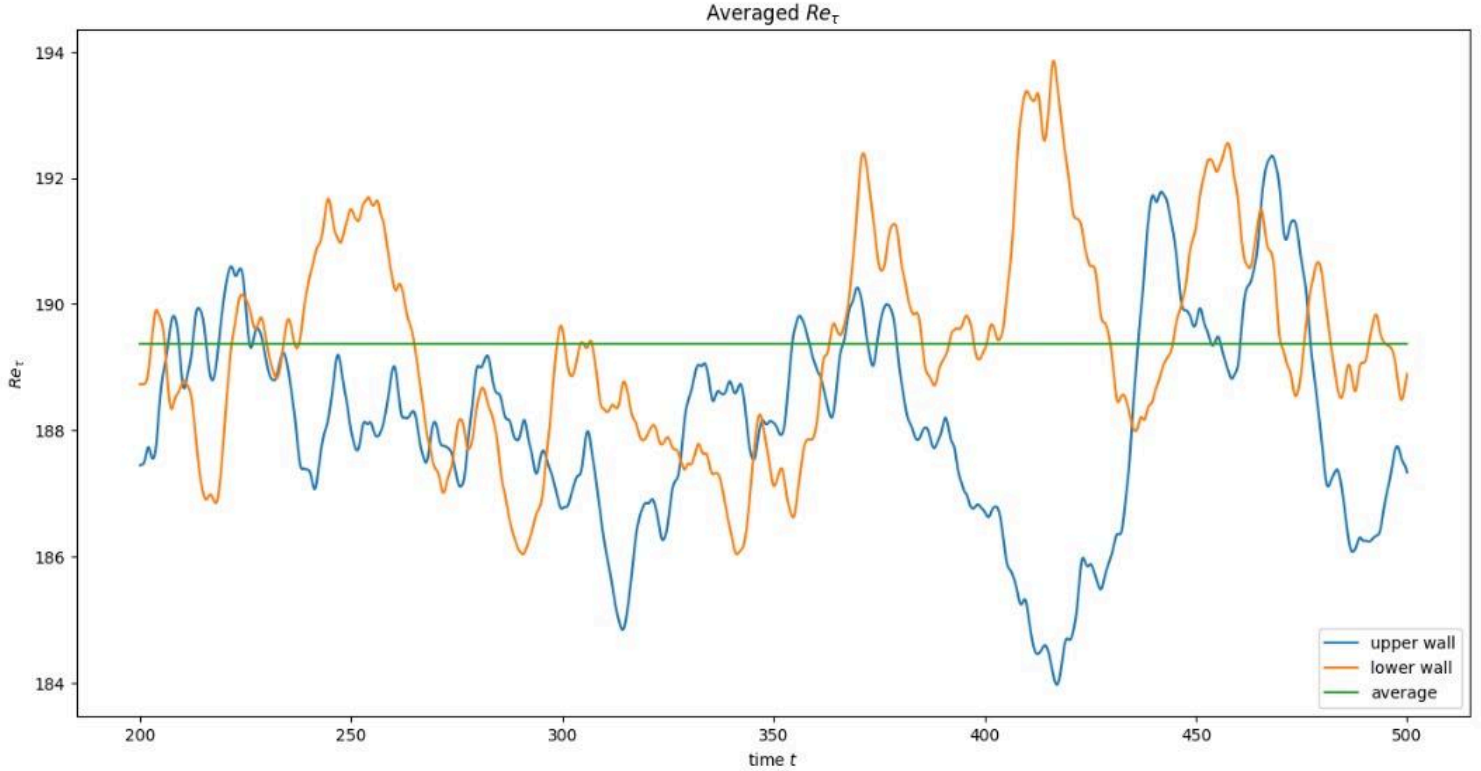
For $Re = 4200$ and $|du/dy|$ is extracted from the file amp.stat , and we know that

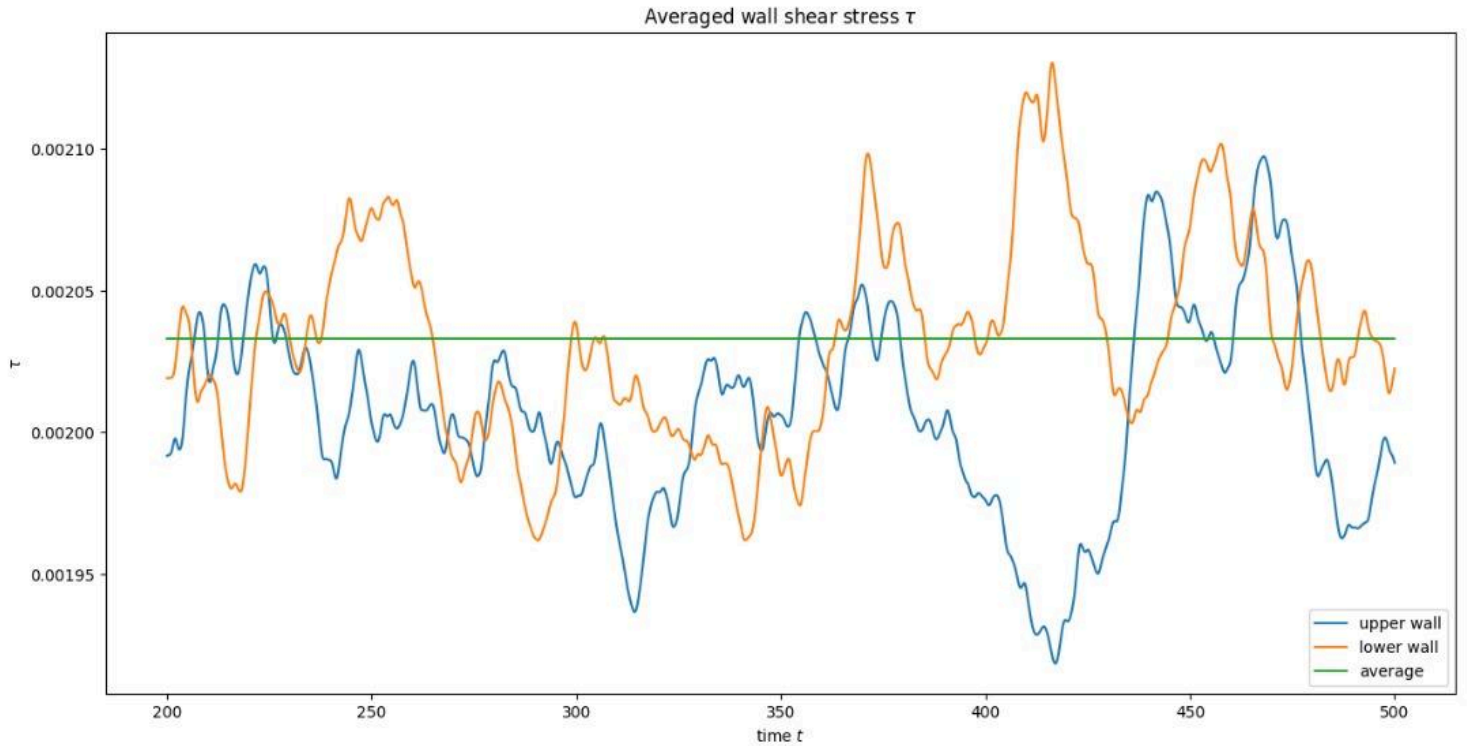
$$T_{w_1} = \nu \left| \frac{du}{dy} \right|_{\text{Top Wall}} \quad T_{w_2} = \nu \left| \frac{du}{dy} \right|_{\text{Bottom Wall}}$$

Average wall shear stress

$$T_w = \frac{T_{w_1} + T_{w_2}}{2}$$

The computed values for averaged friction reynolds number $Re_{\tau} = 189.37018$ and for the averaged wall shear stress $\tau = 0.00203$.





Task 5:

The autocorrelation helps us determine how the signal (wall shear stress in this case) would change if compared to itself at zero lag. A correlation of 1 shows strongly related values and -1 shows strong anti-correlation (we have run the simulation again with a constant Δt of 0.02 to make calculations easier). The transients $t \leq 200$ were discarded. We see that only a short-term correlation exists between turbulent wall shear stress values and it reduces as we increase lag. The formula for autocorrelation is given as:

$$\rho(s) = \frac{\langle q'(t) \cdot q'(t+s) \rangle}{\langle q'(t) \cdot q'(t) \rangle} \quad \text{where } q'(t) \text{ represents the time derivative of lag free signal, } s \text{ represents the lag.}$$

Therefore, the denominator is the normalization with the zero-lag signal. This essentially means how much the signal is changing with respect to itself if it

were shifted by the lag s . The clear negative lobe region is also seen which is similar to what is observed in some literatures [DOI:[10.1063/1.5028324](https://doi.org/10.1063/1.5028324)]

